

Capture and analyse the packets:

<http://gaia.cs.umass.edu/wireshark-labs/HTTPwireshark-file1.html>

- Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

The image shows a Wireshark packet capture of an HTTP GET request. The packet list at the top shows several packets, with packet 351 (GET /wireshark-labs/HTTPwireshark-file1.html HTTP/1.1) selected. The packet details pane on the left shows the structure of the packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane on the right shows the raw data of the packet.

Frame 351: 525 bytes on wire (4200 bits), 525 bytes captured (4200 bits) on interface \Device\NPF...
> Ethernet II, Src: CloudNet_3f:c3:0b (30:03:c8:3f:c3:0b), Dst: Broadcom_1e:5f:a0 (08:0a:f7:1e:5f:a0)
> Internet Protocol Version 4, Src: 192.168.111.50, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 61031, Dst Port: 80, Seq: 1, Ack: 1, Len: 471
Hypertext Transfer Protocol
GET /wireshark-labs/HTTPwireshark-file1.html HTTP/1.1
[Expert Info (Chat/Sequence): GET /wireshark-labs/HTTPwireshark-file1.html HTTP/1.1
Request Method: GET
Request URI: /wireshark-labs/HTTPwireshark-file1.html
Request Version: HTTP/1.1
Host: gaia.cs.umass.edu
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/10.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTPwireshark-file1.html]

- What languages (if any) does your browser indicate that it can accept to the server?

The image shows the details of the selected HTTP GET request. The Hypertext Transfer Protocol section is expanded, showing the request method, URI, version, host, connection, upgrade-insecure-requests, user-agent, accept, accept-encoding, and accept-language headers.

Transmission Control Protocol, Src Port: 61031, Dst Port: 80, Seq: 1, Ack: 1, Len: 471
Hypertext Transfer Protocol
GET /wireshark-labs/HTTPwireshark-file1.html HTTP/1.1
[Expert Info (Chat/Sequence): GET /wireshark-labs/HTTPwireshark-file1.html HTTP/1.1
Request Method: GET
Request URI: /wireshark-labs/HTTPwireshark-file1.html
Request Version: HTTP/1.1
Host: gaia.cs.umass.edu
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/10.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.9
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTPwireshark-file1.html]
[HTTP request 1/2]
[Response in frame: 386]
[Next request in frame: 403]

- What is the IP address of your computer? Of the gaia.cs.umass.edu server?

```
> Frame 351: 525 bytes on wire (4200 bits), 525 bytes captured (4200 bits) on interface \Device\NPF_{6C
> Ethernet II, Src: CloudNet_3f:c3:0b (30:03:c8:3f:c3:0b), Dst: Broadcom_1e:5f:a0 (00:0a:f7:1e:5f:a0)
> Internet Protocol Version 4, Src: 192.168.111.50, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 61031, Dst Port: 80, Seq: 1, Ack: 1, Len: 471
v Hypertext Transfer Protocol
  v GET /wireshark-labs/HTTPwireshark-file1.html HTTP/1.1\r\n
```

- What is the status code returned from the server to your browser?

```
v Hypertext Transfer Protocol
  v HTTP/1.1 200 OK\r\n
    > [Expert Info (Chat/Sequence): HTTP/1.1 200 OK\r\n]
      Response Version: HTTP/1.1
      Status Code: 200
      [Status Code Description: OK]
      Response Phrase: OK
```

- When the HTML file that you are retrieving was last modified at the server?

```
v Hypertext Transfer Protocol
  v HTTP/1.1 200 OK\r\n
    Date: Thu, 19 Oct 2023 12:42:15 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33
    Last-Modified: Thu, 19 Oct 2023 05:59:02 GMT\r\n
    ETag: "20c7-6080b73e23710"\r\n
```

- How many bytes of content are being returned to your browser?

<http://gaia.cs.umass.edu/wireshark-labs/HTTPwireshark-file4.html>

- Analyse the packet and check whether any image file is there? If so, save the file in your local folder and display it.
- Check any packet's status is given as 'Not Modified'. Why the packet is showing the above status.